Boosting and concentration of measure methods in Machine Learning

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Projects at Centro Pi

My thesis is result of three applications of Machine Learning to real industrial problems

- Stone Pagamentos (2020): Credit Scoring
  - ExactBoost: directly boosting the margin in combinatorial and non-decomposable metrics
- Dasa (2021): Uncertainty Quantification
  - Split conformal prediction for dependent data
- Rede Globo (2022): Record Linkage
  - Learning to hash via boosting

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Learning to hash via boosting

Learning to hash via boosting

How to match similar entries from possibly large datasets?

Learning to hash via boosting

Duplicated entries in the same dataset

Restaurants			
name	address	location	cuisine
arnie morton's of chicago	435 s. la cienega blv.	los angeles	american
nate 'n' al's	414 n. beverly dr.	los angeles	american
⋮	⋮	⋮	⋮
schatzi on main	3110 main st.	los angeles	continental
arnie morton's of chicago	435 s. la cienega blvd.	los angeles	steakhouses

Learning to hash via boosting

Duplicated entries in different datasets

People (A)				
last_name	last_name	birth_month	birth_day	full_name
GERD	BAUER	7	27	GERD BAUER
⋮	⋮	⋮	⋮	⋮
WOLFGANG	ENGEL	12	27	WOLFGANG ENGEL

People (B)			
last_name	last_name	birth_year	full_name
HARALD	WEBER	1977	HARALD WEBER
⋮	⋮	⋮	⋮
GERD	BAUER	1968	GERD BAUER

Learning to hash via boosting

Matching datasets can reduce data size

Restaurants			
name	address	location	cuisine
arnie morton's of chicago	435 s. la cienega blv.	los angeles	american
⋮	⋮	⋮	⋮
arnie morton's of chicago	435 s. la cienega blvd.	los angeles	steakhouses

Matched Restaurants			
name	address	location	cuisine
⋮	⋮	⋮	⋮
arnie morton's of chicago	435 s. la cienega blvd.	los angeles	[american, steakhouse]

Matching datasets can improve data quality

People (A)				
last_name	last_name	birth_month	birth_day	full_name
GERD	BAUER	7	27	GERD BAUER
⋮	⋮	⋮	⋮	⋮

People (B)			
last_name	last_name	birth_year	full_name
⋮	⋮	⋮	⋮
GERD	BAUER	1968	GERD BAUER

Linked People (A+B)					
last_name	last_name	birth_year	birth_month	birth_day	full_name
GERD	BAUER	1968	7	27	GERD BAUER
⋮	⋮	⋮	⋮	⋮	⋮

Learning to hash via boosting

- **Naive solution:**
  - Check all pairs for similarity
  - Prohibitively expensive
- **Hashing solution:**
  - Construct a low-dimension binary representation for each entry
  - Only similar entries should have same code

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  - Prohibitively expensive
- **Hashing solution:**
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  - Only similar entries should have same code

Learning to hash via boosting

Restaurants				
name	address	location	cuisine	hash
arnie morton's of chicago	435 s. la cienega blv.	los angeles	american	010001
nate 'n' al's	414 n. beverly dr.	los angeles	american	000011
⋮	⋮	⋮	⋮	⋮
schatzi on main	3110 main st.	los angeles	continental	110101
arnie morton's of chicago	435 s. la cienega blvd.	los angeles	steakhouses	010001

- Performance metrics for hashing solutions:
  - **Recall:** proportion of similar pairs that are matched
  - **Reduction Ratio (RR):** proportion of pairwise comparisons avoided

Learning to hash via boosting

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name	address	location	cuisine	hash
arnie morton's of chicago	435 s. la cienega blv.	los angeles	american	010001
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Learning to hash via boosting

Existing approaches for hashing:

- Manually select and tune the hashing method
- Unsupervised schema-agnostic approaches (e.g. TLSH[1])
- Learning to hash
  - Kernelized LSH (KLSH)[2]
  - Blocking Schemes Learners (BSL) [3]
  - Deep Learning models[4]
- Usually methods are either data-independent (TLSH), computationally intensive (DL models, KLSH) or lack of theoretical bounds (BSL, DL models)

[1] Rebecca C. Steorts, Samuel L. Ventura, Mauricio Sadinle, Stephen E. Fienberg, "A Comparison of Blocking Methods for Record Linkage"

[2] Brian Kulis; Kristen Grauman, "Kernelized locality-sensitive hashing for scalable image search"

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Learning to hash via boosting

Our main contributions:

- We propose a Learning to hash, data-dependent algorithm based on Adaboost[5]
  - Adaboost is a classification model but we adapt it to create a similarity function between entries
- The algorithm is fast and easy to implement
- Theoretical bounds based on boosting and concentration of measure theory to control Recall and Reduction Ratio (RR) metrics

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Learning to hash via boosting

Our model follows two steps:

- **Step 1:** Learn a similarity function from the data
- **Step 2:** Use this similarity function to construct the hashes

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Learning to hash via boosting

Step 1: The model learns simple similarity rules from the data

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- Rule 1: Name starts with "a"
- Rule 2: Is in Los Angeles
- Rule 3: Is steakhouse cuisine

⋮

- Rule T: The second letter in the address is "l"

Learning to hash via boosting

Step 1: The model learns simple similarity rules from the data

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- $\mathbf{Rule}_1$: Name starts with "a"?
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$\vdots$

- $\mathbf{Rule}_T$: The second letter in its name is a "r"?

Learning to hash via boosting

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Learning to hash via boosting

Step 1: For each Rule, the model learns positive weights associated to its relevance and an error value

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- $\mathbf{Rule}_1$ has relevance $\alpha_1 = 0.31$ and $\text{error}_1 = 0.10$
- $\mathbf{Rule}_2$ has relevance $\alpha_2 = 0.29$ and $\text{error}_2 = 0.15$
- $\mathbf{Rule}_3$ has relevance $\alpha_3 = 0.25$ and $\text{error}_3 = 0.20$

$\vdots$

- $\mathbf{Rule}_T$ has relevance $\alpha_T = 0.15$ and $\text{error}_T = 0.40$

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$\vdots$

- $\mathbf{Rule}_T$ has relevance $\alpha_T = 0.15$ and $\text{error}_T = 0.40$

Learning to hash via boosting

Step 1: Using such rules and its weights, we can construct a similarity function between items A and B given by:

$$f^*(A, B) = \sum_{i=1}^{T} \alpha_i \mathrm{Rule}_i(A) \mathrm{Rule}_i(B)$$

where $\mathrm{Rule}_i(x)$ is 1 if x satisfies Rule_i and -1 otherwise

Learning to hash via boosting

Desideratum (What we want!)

Let a pair of entries (A, B) and define $y = 1$ if A and B are a match, and $y = -1$ otherwise. Given Rules $(Rule)_{t=1}^{T}$ and convex weights α_t, let

$$f^*(A, B) = \sum_{i=1}^{T} \alpha_i Rule_i(A) Rule_i(B).$$

Then, for a given $\theta > 0$, the following holds with probability at least $1 - \varepsilon$ over the choice of $(A, B, y) \sim P$,

$$y\, f^*(A, B) > \theta, \tag{1}$$

Intuitively,

- if A and B are match then $f^*(A, B) > \theta$, that is, very positive
- if A and B are not match then $f^*(A, B) < -\theta$, that is, very negative

Learning to hash via boosting

Theorem (Theorem 3.3 from thesis)

If $\theta > 0$ and $\delta > 0$ are given, then f^* satisfies the previous condition with probability at least $1 - \delta$, with the value of $\varepsilon > 0$ given by:

$$\varepsilon := 2^T \prod_{t=1}^{T} \mathrm{error}_t^{1/2-\theta} (1 - \mathrm{error}_t)^{\theta - 1/2} \tag{2}$$

$$+ \frac{8}{\theta} \left(\mathcal{R}_{\mathcal{S}_{\mathcal{A},n}}(\mathcal{K}) + \mathcal{R}_{\mathcal{S}_{\mathcal{B},n}}(\mathcal{K}) \right) + \sqrt{\frac{\log(1/\delta)}{2n}}. \tag{3}$$

Furthermore, if there exists $\gamma > 0$ such that for all $t \in [T]$, $\gamma \leq (1/2 - \mathrm{error}_t)$ and $\theta \leq 2\gamma$, then the term in (2) decreases exponentially with T.

The proof relies on concentration of measure, margin theory and Rademacher complexity properties

Learning to hash via boosting

Step 2: Given rules Rule_t, weights $\alpha_t > 0$ and the similarity function f^* from the previous step

- To construct a single-bit hash we draw a random rule R following the distribution $\mathbb{P}\left(R = \mathrm{Rule}_i\right) = \alpha_i$
- We say that two items A and B have the same single-bit hash if $R(A) = R(B)$
- It is easy to show that

$$\mathbb{P}\left[R(A) = R(B)\right] = \frac{1 + f^*(A, B)}{2}.$$

- We concatenate several single-bit hashes to construct a final hash function

Learning to hash via boosting

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$$\mathbb{P}\left[R(A) = R(B)\right] = \frac{1 + f^*(A, B)}{2}.$$

- We concatenate several single-bit hashes to construct a final hash function

Learning to hash via boosting

We combine several single-bit hashes via the following algorithm:

Algorithm Algorithm to construct the hash codes

Require: $k, L \in \mathbb{N}$, convex weights$(\alpha_t)_{t=1}^T$, Rules $(\mathrm{Rule}_t)_{t=1}^T$

```
for i ← 1 to L do
    for j ← 1 to k do
        g_{i,j} ← Rule_t with probability α_t
    end for
    g_i ← (g_{i,1}, ..., g_{i,k})
end for
g ← (g_1, ..., g_L)
return g
```

Two elements A and B are tested for similarity if $g_i(A) = g_i(B)$ for some $i = 1, \ldots, L$

Learning to hash via boosting

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Theorem (Theorem 3.4 from thesis)

Consider datasets $\mathcal{A}$ and $\mathcal{B}$ such that $|\mathcal{A}| = N_{\mathcal{A}}$ and $|\mathcal{B}| = N_{\mathcal{B}}$. Suppose our condition holds for $\theta > 0$. Then, given $\gamma \in (0,1)$, if we set:

$$\rho := \frac{\log\left(\frac{2}{1+\theta}\right)}{\log\left(\frac{2}{1-\theta}\right)}, \; k := \lceil \log_{\frac{2}{1+\theta}} N_{\mathcal{A}} \cdot N_{\mathcal{B}} \rceil \text{ and } L := \left\lceil \frac{2(N_{\mathcal{A}} \cdot N_{\mathcal{B}})^{\rho} \log(1/\gamma)}{1+\theta} \right\rceil,$$

then

$$\mathbb{E}\left[\text{Recall}\right] \geq (1-\gamma)(1-\varepsilon), \quad \mathbb{E}\left[\text{RR}\right] \geq \left(1 - \frac{|\mathcal{M}| + L}{N_{\mathcal{A}} \cdot N_{\mathcal{B}}}\right)(1-\varepsilon).$$

Both expectations are with respect to the randomness in the hash code.

Learning to hash via boosting

Our model is competitive to classical Record Linkage models

RLDATA10000		
model	recall	reduction ratio
our model	**0.9925 ±0.0055**	0.9916 ± 0.0009
TLSH	0.9862 ± 0.0079	**0.9990 ± 0.0003**
KLSH	0.9340 ± 0.0152	0.9872± 0.0004

Restaurant		
model	recall	reduction ratio
our model	0.9296 ±0.07.03	0.9800 ± 0.0046
TLSH	0.7977 ± 0.0862	**0.9903 ± 0.0056**
KLSH	**0.9872 ± 0.0284**	0.9869± 0.0017

Conformal prediction for dependent data

Conformal prediction[6]

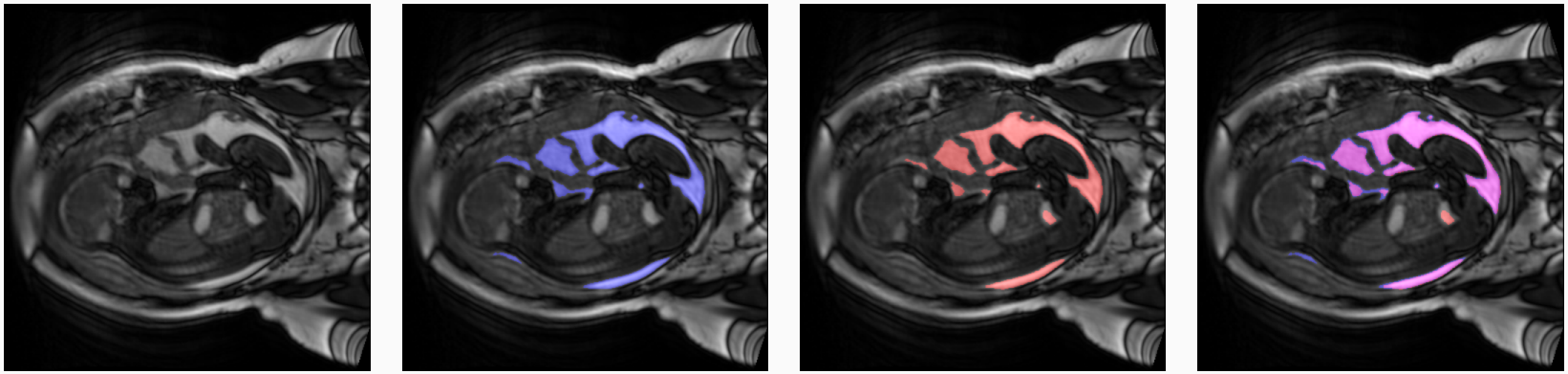
[6] Vladimir Vovk, Alexander Gammerman, Glenn Shafer, “ Algorithmic Learning in a Random World ”

Conformal prediction

How to construct predictive intervals for general black-box algorithms?

Conformal prediction

Amniotic fluid prediction using U-net[7]



(a) Input (b) Target (c) Prediction (d) Evaluation

We can derive the volume of AF from the segmented image

[7] Daniel Csillag, Lucas Monteiro, Thiago Ramos, João Vitor Romano, Rodrigo Schuller, Roberto B. Seixas, Roberto I. Oliveira, Paulo Orenstein, "AmnioML: Amniotic Fluid Segmentation and Volume Prediction with Uncertainty Quantification"

Conformal prediction

- Usually, precise predictions made by neural nets
- It is important to quantify the uncertainty of such methods

  - Prediction is 2.31 but
    - I'm 90% sure the true AE volume is between 1.62 and 3.01
    - I'm 99% sure the true AE volume is between 1.04 and 7.01

Conformal prediction

- Usually, precise predictions made by neural nets
- It is important to quantify the uncertainty of such methods

Prediction is 2.8L but:

- I'm 90% sure the true AF volume is between 2.72L and 2.88L
- I'm 90% sure the true AF volume is between 1.90L and 3.70L

Conformal prediction

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  Prediction is 2.8L but:

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Conformal prediction

- Given data $\{(X_i, y_i)\}_{i=1}^n$ and $\alpha \in (0,1)$, can we construct a prediction set $C_{1-\alpha}$ such that, for a new point (X_{n+1}, y_{n+1})

$$\mathbb{P}\left[y_{n+1} \in C_{1-\alpha}(X_{n+1})\right] \geq 1 - \alpha?$$

  For example, X_i could be an input exam and y_i its AF volume
- If the data is exchangeable, we can use Split Conformal Prediction[8]
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Conformal prediction

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Conformal prediction

Split Conformal Prediction (SCP) procedure:

- Split the data in train, calibration sizes n_{train} and n_{cal}, respectively
- Train the prediction model $\widehat{\mu}$
- Apply score $\widehat{s}(x, y) \in \mathbb{R}$ over calibration set measuring how far the prediction is from the true value

$$\text{e.g.} \quad \widehat{s}(x, y) = |\widehat{\mu}(x) - y|$$

- Define

$$C_{1-\alpha}(x) = \{y : \widehat{s}(x, y) \leq \widehat{q}_{(1+n_{\text{cal}})(1-\alpha)}\}$$

where

$$\widehat{q}_{1-\alpha} := \arg\min_{t \in \mathbb{R}} \left\{ \frac{1}{n_{\text{cal}}} \sum_{(x,y) \in \text{cal}} \mathbb{1}[\widehat{s}(x, y) \leq t] \geq 1 - \alpha \right\}$$

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Conformal prediction

Theorem (Marginal Coverage)

Given exchangeable data $\{(X_i, y_i)\}_{i=1}^n$ *and* $\alpha \in (0,1)$*, let* $C_{1-\alpha}$ *as in SCP procedure. Then, for any new point* (X_{n+1}, y_{n+1})

$$\mathbb{P}\left[y_{n+1} \in C_{1-\alpha}(X_{n+1})\right] \geq 1 - \alpha.$$

Additionally, if $\widehat{s}_j$ *are almost surely distinct, then*

$$\mathbb{P}\left[y_{n+1} \in C_{1-\alpha}(X_{n+1})\right] \leq 1 - \alpha + 1/(n_{\text{cal}} + 1).$$

Conformal prediction

- Pros:
  - SCP provides valid coverage and finite-sample guarantees
  - Algorithm is fast and easy to implement
  - Works for any model $\widehat{\mu}$ irrespective of its performance
  - Works for arbitrary discrepancy scores

$$\widehat{s}(x,y) = |\widehat{\mu}(x) - y| \quad \text{or} \quad \widehat{s}(x,y) = \frac{|\widehat{\mu}(x) - y|}{\widehat{\mu}(x)}$$

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Our main contributions:

- General framework that extended Split Conformal Prediction to a large family of process
- Finite sample theoretical guarantees for based on concentration of measure theory
- Experiments and applications to back up our theoretical claims

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Theorem (Theorem 4.1 from thesis)

Under certain ***decoupling and concentration of measure conditions*** *there exists small* $\eta > 0$*, such that, given* $\alpha \in (0, 1)$*, for any new point* (X_{n+1}, y_{n+1})*:*

$$\mathbb{P}[y_{n+1} \in C_{1-\alpha}(X_{n+1})] \geq 1 - \alpha - \eta.$$

Additionally, if $\widehat{s}(X, Y)$ *almost surely has a continuous distribution conditionally on the training data, then*

$$|\mathbb{P}[Y_{n+1} \in C_{1-\alpha}(X_{n+1})] - (1 - \alpha)| \leq \eta.$$

Conformal prediction

Such decoupling and concentration of measure conditions are satisfied by large classes of processes

- Usual Split Conformal for i.i.d. data
  - Marginal, conditional and empirical guarantees
- Risk-controlling prediction sets[9]
- Non-stationary data[10]
- Rank-one-out conformal prediction [11]
- Stationary β-mixing processes (dependent data)

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Conformal prediction for dependent data

Stationary β-mixing scenario:

- Stationarity:

$$(Z_t, \ldots, Z_{t+m}) \stackrel{d}{=} (Z_{t+k}, \ldots, Z_{t+m+k}) = Z_{(t+k):(t+m+k)}.$$

- β-mixing:

$$\beta(a) = \|\mathbb{P}_{-\infty:0,a:\infty} - \mathbb{P}_{-\infty:0} \otimes \mathbb{P}_{a:\infty}\|_{\mathrm{TV}} \to 0.$$

Intuitively, $\mathbb{P}_{-\infty:0,a:\infty}$ is like $\mathbb{P}_{-\infty:0} \otimes \mathbb{P}_{a:\infty}$ up to a factor of $\beta(a)$

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Theorem (Marginal coverage: stationary β-mixing processes)

Suppose that $(X_i, Y_i)_{i=1}^n$ is stationary β-mixing. Then given $\alpha \in (0,1)$ and $\delta > 0$, there exists small $\eta > 0$ such that, for any new point (X_{n+1}, y_{n+1}):

$$\mathbb{P}\left[y_{n+1} \in C_{1-\alpha}(X_{n+1})\right] \geq 1 - \alpha - \eta,$$

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$\eta = \varepsilon_{\text{cal}} + \varepsilon_{\text{train}} + \delta$, where $\varepsilon_{\text{train}} = \beta(n + 1 - n_{\text{train}})$ and

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Idea of the proof:

- Transform stationary β-mixing problem into a i.i.d. problem
- Apply Bernstein's Inequality for β-mixing processes
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- Let $\{Z_i\}_{i=1}^n$ be stationary β-mixing with $|Z_i| \leq 1$. The sequence

$$Z_1, Z_2, Z_3, Z_4, Z_5, Z_6, Z_7, Z_8, \ldots, Z_n$$

  is not i.i.d..
- Up to a factor of $\beta(3)$ we can assume that (Z_1, Z_2) is independent of $(Z_6, Z_7, \ldots)$

$$Z_1, Z_2, Z_3, Z_4, Z_5, Z_6, Z_7, Z_8, Z_9 \ldots$$

- Repeating this process, **up to a factor depending on** β, the blue and red sequences are i.i.d.[12]

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[12] Bin Yu., "Rates of Convergence for Empirical Processes of Stationary Mixing Sequences"

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Lemma (Bernstein's inequality for stationary β-mixing process)

Let $Z_1, \ldots, Z_n$ be a sample drawn from a stationary β-mixing distribution with $Z_1 \in [0,1]$ and $\mathrm{Var}[Z_1] = v < \infty$. Then, for any $m, a, s \in \mathbb{N}_+$ with $m > 1$, $n = 2ma + s$ and $\delta > 4(m-1)\beta(a))$, with probability at least $1 - \delta$ it holds that

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$$\widetilde{\sigma}(a) = \sqrt{\frac{1}{4} + \frac{2}{a}\sum_{j=1}^{a-1}(a-j)\beta(j)}.$$

Conformal prediction for dependent data

$\eta = \varepsilon_{\text{cal}} + \varepsilon_{\text{train}} + \delta$, where $\varepsilon_{\text{train}} = \beta(n+1-n_{\text{train}})$ and

$$\varepsilon_{\text{cal}} := \inf_{(a,m,r)\in F_{\text{cal}}} \left\{ \widetilde{\sigma}(a)\sqrt{\frac{4}{n_{\text{cal}}-r+1}\log\left(\frac{4}{\delta_{\text{cal}}-4(m-1)\beta(a)-\beta(r)}\right)} + \frac{1}{3m}\log\left(\frac{4}{\delta_{\text{cal}}-4(m-1)\beta(a)-\beta(r)}\right) + \frac{r-1}{n_{\text{cal}}} \right\},$$

with

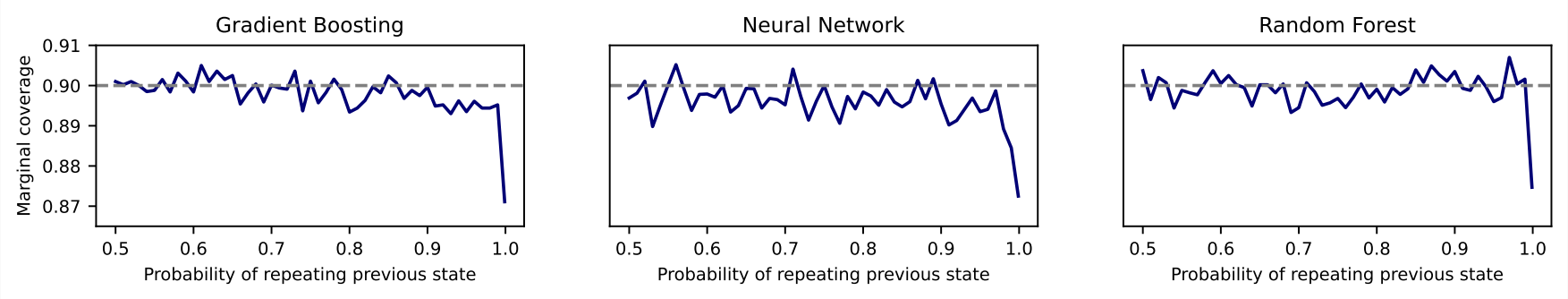
$$F_{\text{cal}} = \{(a,m,r)\in\mathbb{N}^3_{>0} \;:\; 2ma = n_{\text{cal}}-r+1,\; \delta_{\text{cal}} > 4(m-1)\beta(a)+\beta(r)\}$$

and

$$\widetilde{\sigma}(a) = \sqrt{\frac{1}{4} + \frac{2}{a}\sum_{j=1}^{a-1}(a-j)\beta(j)}.$$

Conformal prediction for dependent data

Artificial data: two-state hidden Markov model



How marginal coverage calculated through $10\,000$ simulations, is affected by increasing levels of dependence, x-axis, for three different models (boosting, neural network and random forest) in and $n_{\text{train}} = 1000, n_{\text{cal}} = 500$

Conclusion

Knowing classical Machine Learning theory was fundamental, not only for the theoretical part, but also to guide us to create new practical and efficient algorithms